



WIESŁAW PIETRUSZKIEWICZ

Integration with the external systems

In-house solutions vs outsourcing – in general

In-house

- Full control of process
- Flexibility
- No external risk factors (failures, dependency etc.)
- Customised to particular needs
- Gained experience
- Potential to grow

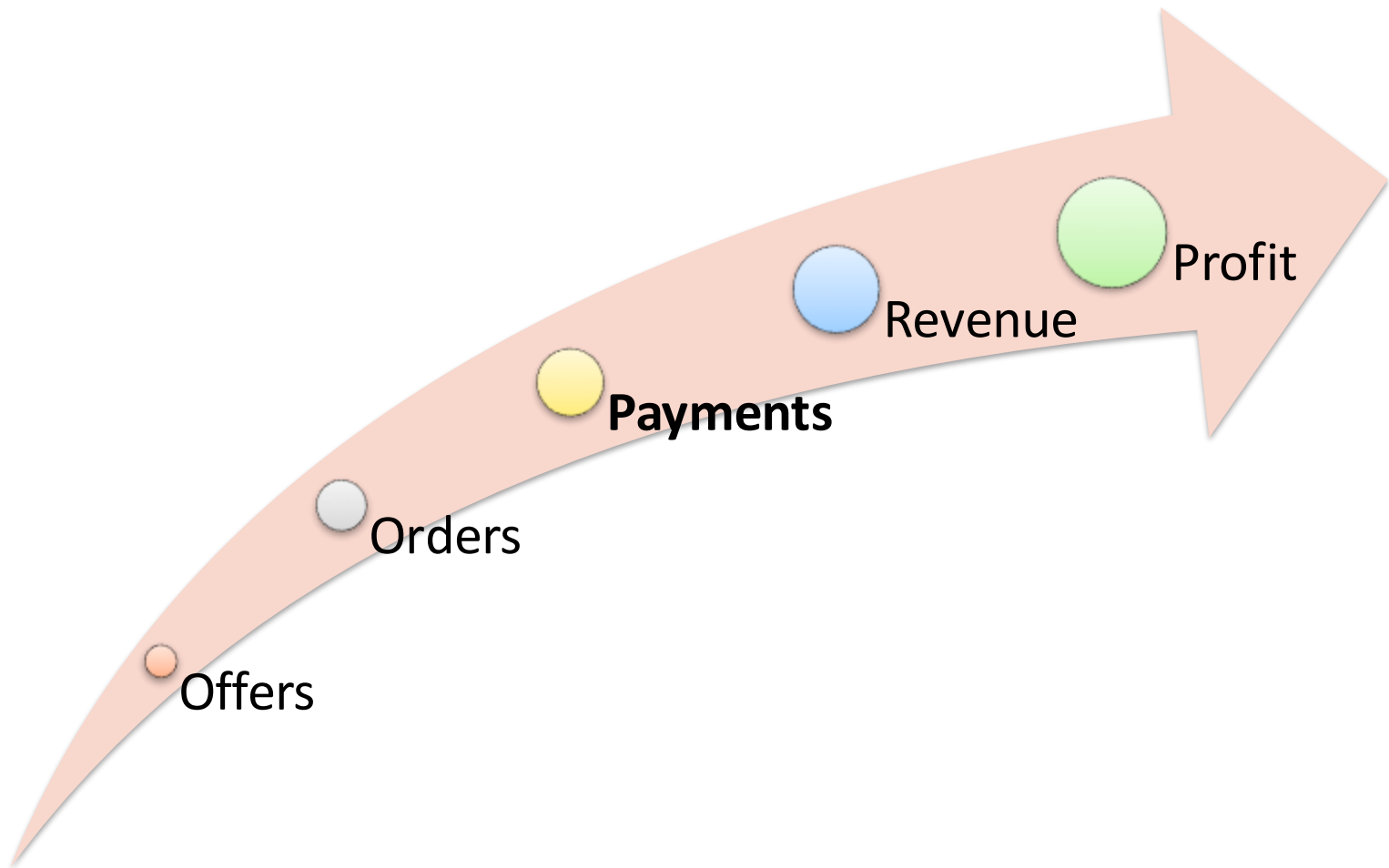
Outsourcing

- Easiness of usage (relative)
- Effects of scale
- Time of deployment
- Quality of services
- Customisation limited by external constraints
- Efficient usage of resources
- Focus on key business processes

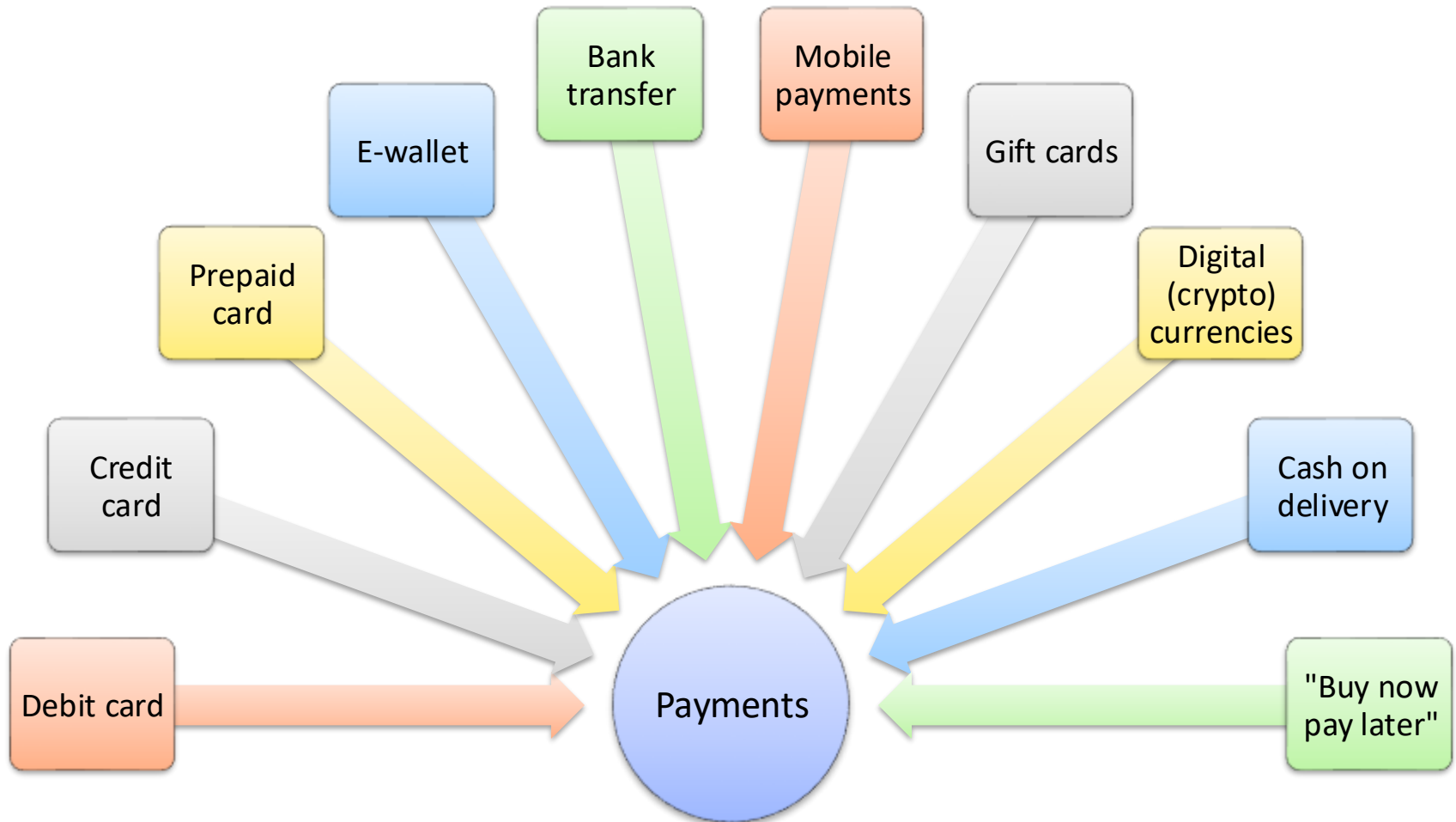
Integration with the external systems

PAYMENTS

Payments – The essentials



Payments – Types



Debit card

- Debit card is issued by a bank, typically in one of two major systems (Visa or Mastercard)
- Card is linked with an account
- Payments are done against the account
- Popular in developed countries
- Widely trusted
- Common for online and offline transactions
- Risk of frauds
- Limited competition and (thus) transaction fees

Credit card

- Partially similar to debit card
- Credit card may be issued by a non-bank
- Credit card isn't linked with a banking account
- Payments have usually a monthly limit
- More popular among the older customers
- More popular in the developed countries

Debit & credit cards 🙄

Debit Card Retweeted



dev @devbordurr · 29 Nov 2016
just got a new credit card 🥰🥰



236

36

302



Debit Card Retweeted



DJ K-VO @DjKvo_Oficial · 21 Nov 2016

I have a new credit card! Thank you [@PayoneerEspanol](#) for the support! -
The king, Dj Kvo



75

37

96



Prepaid & virtual cards

- Payments up to the amount kept at the account (prepaid cards) or controlled for each transaction (virtual)
- Safer and more accessible for some customers
- Less convenient due to attention needed



E-wallet

- Payments is done using a wallet linked with a bank account or credit/debit card
- Approval is done by providing a password/pin on a device/application to which e-wallet is connected
- Commonly used on mobile/wearable devices (Apple Pay, Google Pay, Garmin Pay etc.)
- Very fast and convenient for m-commerce

Bank (wire) transfer

- Transfer between the two bank accounts
- Standard for many non-EC payments
- Limited popularity in EC transactions
- No fees for sellers
- Fees for buyers
- Each transfer is authenticated
- Slow (up to a few days)

Direct deposit

- A schema of payments defined by providing a range of data, and used for reoccurring payments (salary, taxes, bills)
- Not commonly accepted in e-commerce
- Diverse popularity in different countries

Mobile (app) payments

- Payments done against an account, managed via mobile app
- Account is linked with other type of payment, or prepaid
- Very fast and convenient
- A fee for prepayments
- Rather used for smaller payments

Gift cards

- A pre-paid card with a set amount of money, that can be redeemed for products or services
- Open-loop (used by any accepting merchant), or closed-loop (to redeemed at the specific e-store – more frequent in EC)
- Increase loyalty and brand awareness
- Bonus – unused money (time-limited validity)
- The purchaser is different from the payee

Digital (crypto) currencies

- Payment done using a cryptocurrency
- Acceptance is changing (see Tesla)
- Very unpredictable exchange rate
- A lot of con companies/people
- Beyond an official control
- Anonymity of both parties
- Popularity on illegal markets
- No protection schema

Cryptocurrencies



Integration with the external systems

Cash on delivery

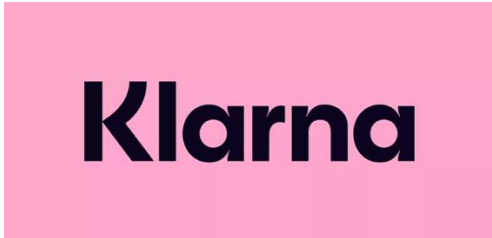
- The traditional way of making a payment
- Limited possibility of a fraud (both parties)
- Not suitable for digital goods
- Suitable for card-less customers
- Requires purchaser to keep cash
- A lag between the time of order and payment

"Buy now pay later"

- A short-term loan, offered during the purchase
- Loans offered by a 3rd party - service provider (they are country-specific e.g. loans companies, fin-techs, mobile companies etc.)
- Gains the popularity (high rate among younger customers)
- No explicit fee
- A high implicit fee

The logo for Afterpay, featuring the word "afterpay" in a bold, black, sans-serif font, followed by a circular icon containing two curved arrows forming a loop. The entire logo is set against a light green rounded rectangular background.

afterpay

The logo for Klarna, featuring the word "Klarna" in a bold, black, sans-serif font, centered within a solid pink rectangular background.

Klarna

QR codes



Integration with the external systems

QR codes – Ideas

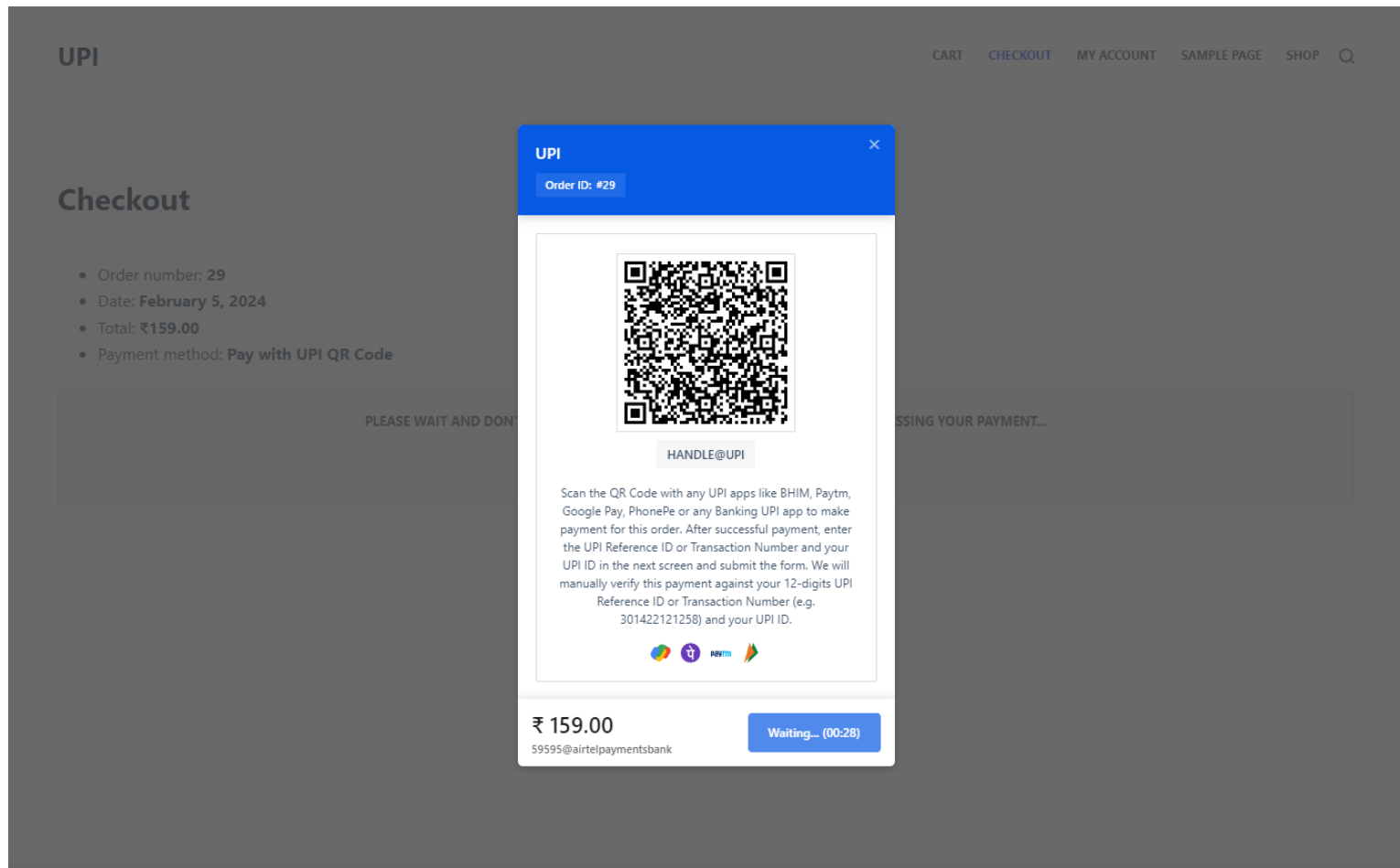
Structure	QR codes are made up of black squares arranged on a white grid, which can be scanned by cameras or QR code readers to retrieve encoded information.
Versatility	They can store various types of data, including URLs , text, contact information, and payment details, making them useful in many contexts.
Ease of Use	Scanning a QR code is quick and straightforward, requiring only a smartphone camera or a dedicated app, which enhances user convenience.
Contactless Interaction	QR codes facilitate contactless transactions and information sharing (COVID-19).
Marketing Tool	Businesses use QR codes in marketing campaigns to direct customers to websites, promotions, or product information, enhancing engagement and tracking.
Error Correction	QR codes have built-in error correction capabilities, allowing them to be scanned even if partially damaged or obscured.
Global Adoption	They are widely used across various industries, including retail, hospitality, healthcare, and logistics, due to their adaptability and efficiency.

Integration with the external systems

QR code payments

- QR code payments allow payments to be done quickly and easily via mobile devices
- A merchant generates a QR code linked to a payment request
- No need for cash or physical cards
- Popular in various regions, especially in Asia
- Many digital wallets and payment platforms (like PayPal, Venmo, and WeChat Pay)

QR code payments in WooCommerce



Payments – Local solutions

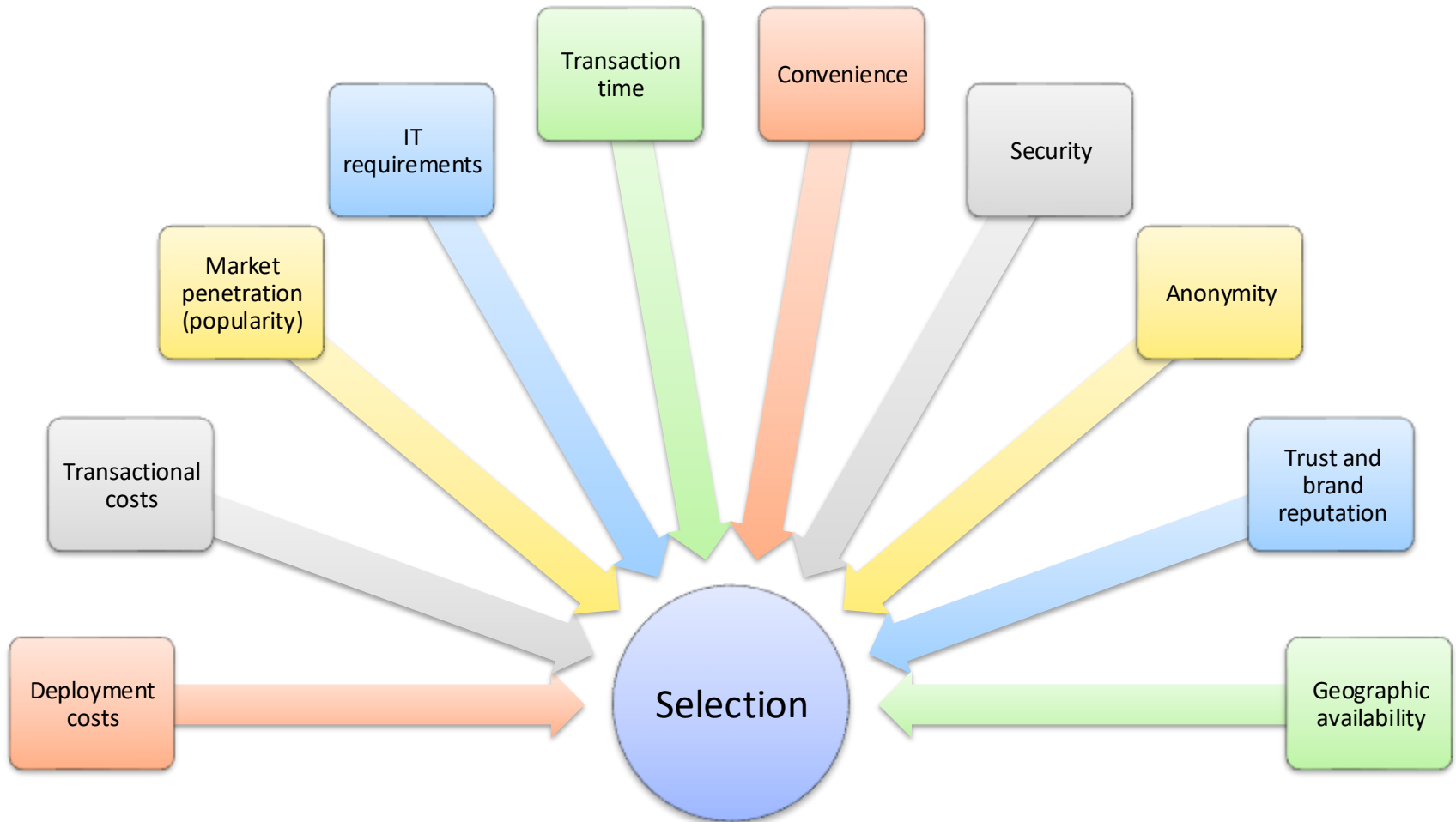


- Created by 6 polish banks
- Payment is in form of a 6 digits
- Generated by a bank application, and valid for 2 minutes
- Payment approved by the owner in bank application
- Different scenarios of payments (also phone → phone)



- Created in Kenya
- Expanded to other countries (mainly African)
- Payment done as SMS
- Used for smaller amounts
- High fees (lack of competition)
- Removal of cash in high-crime areas

Payments – Selection factors



Payments

In-house solutions

- No outsourcing risks
- No 3rd party dependency
- No fees
- Control



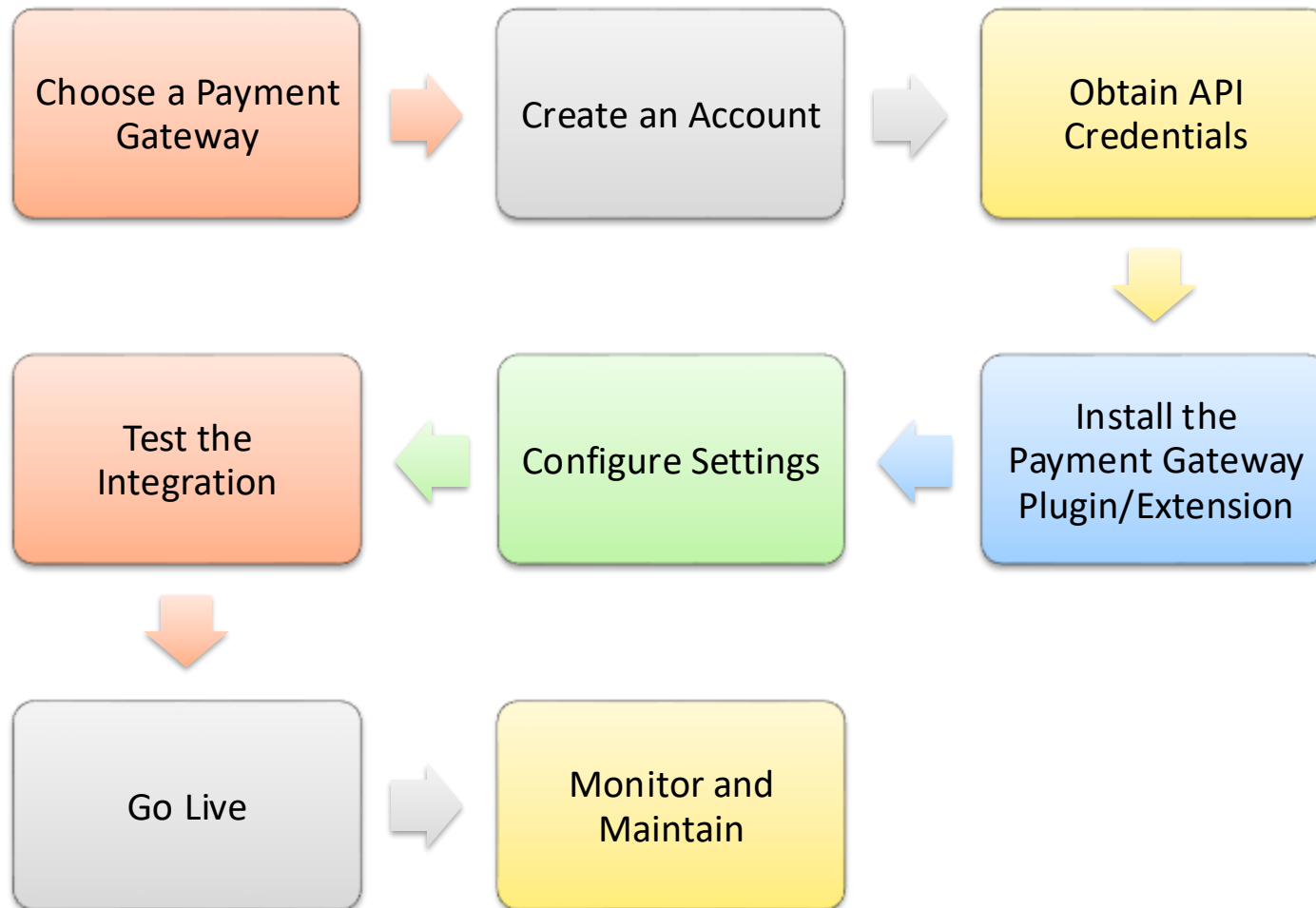
Gateways

- Security
- Versatility
- Cost effectiveness
- Automation
- Add-ons

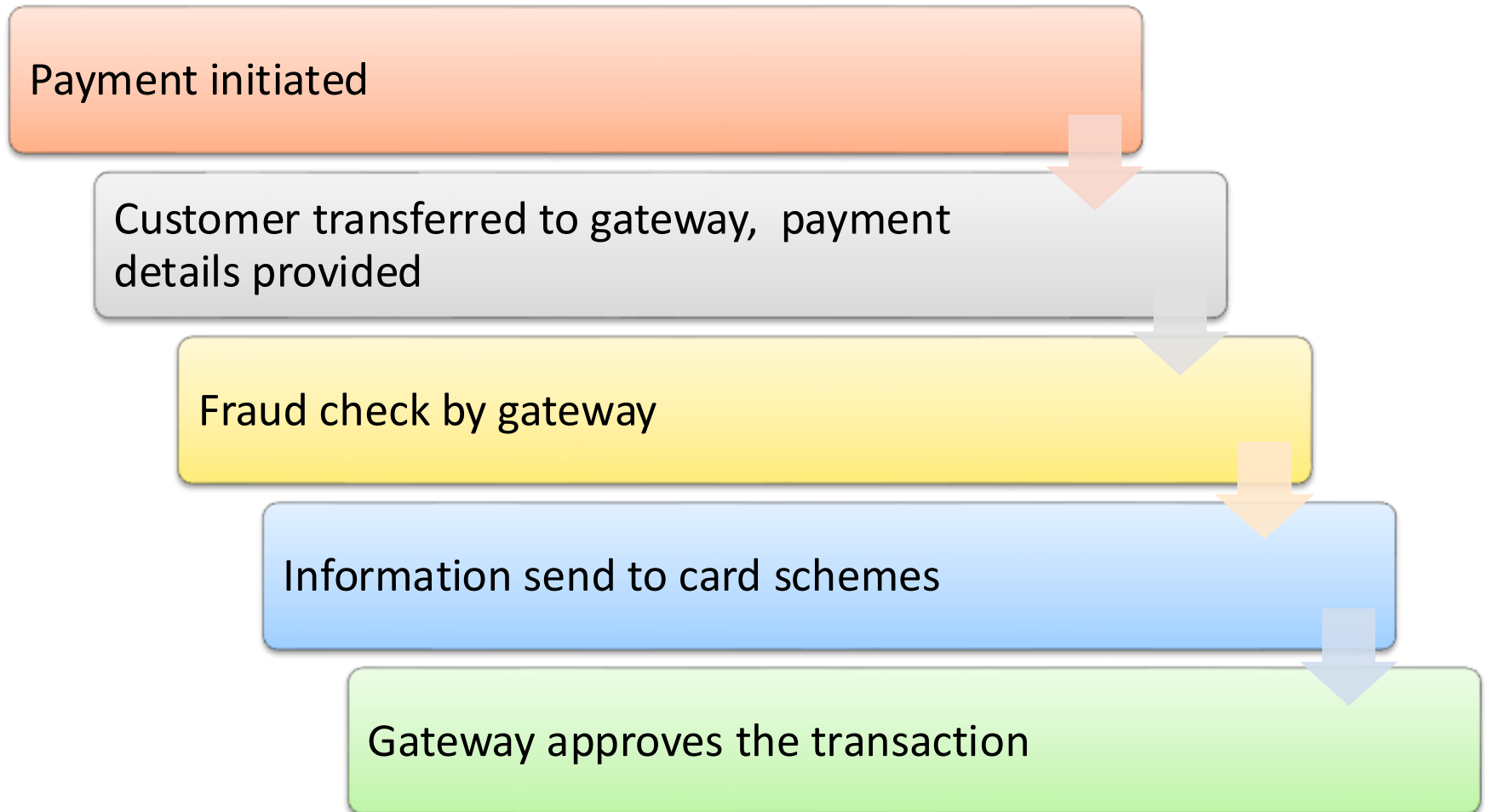


Integration with the external systems

EC integration with payment gateway



Payment via gateways

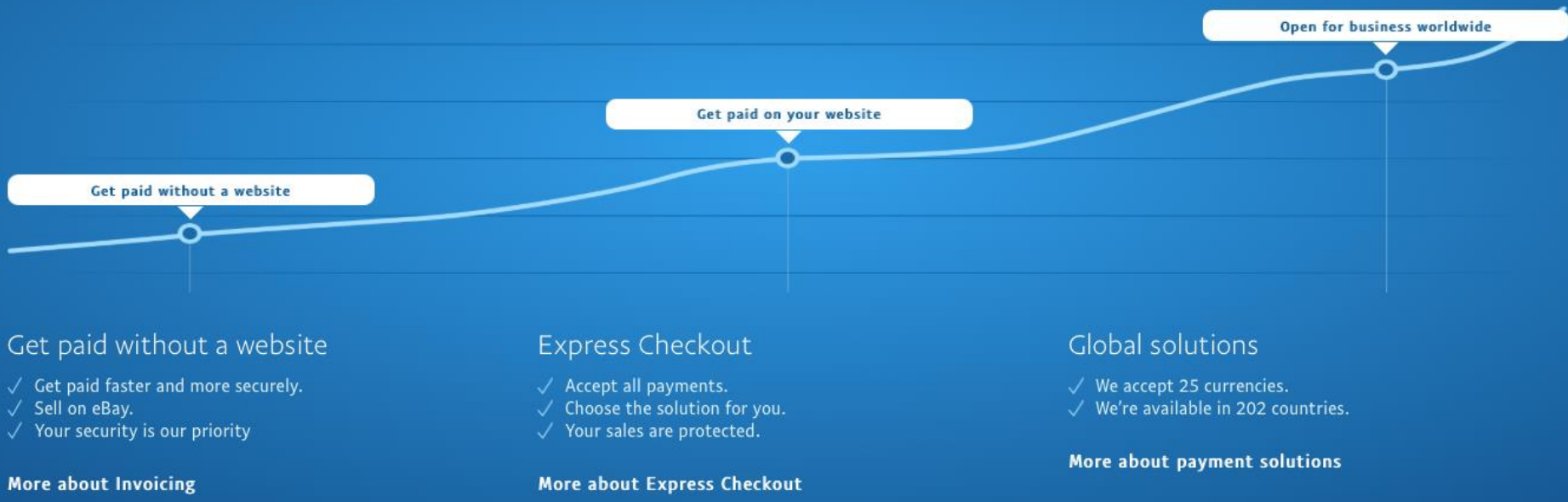


Integration with the external systems

Gateways – PayPal – Services

Grow your business with PayPal.

Start off small and grow, or go big now. We're here to help you boost your sales.



Integration with the external systems

PayPal gateway options

Two gateway options, simple pricing.

PayPal has two gateway options that give you different levels of customization for your online checkout pages.

Compare pricing details

Payflow Link: \$0.00/month

Checkout hosted by PayPal.

At checkout, your customers enter their payment details on a secure, PCI-compliant template hosted by PayPal. You can choose to integrate our embedded template (which sits right in your website), or you can choose a customizable full-page template. All templates include PayPal and PayPal Credit, so your customers have more options to pay.

or

Payflow Pro: \$25.00/month

Unlimited checkout customization.

Payflow Pro is fully customizable, so you can build a checkout experience as unique as your business. Everything is on the table, from language and layout to page sequence and PCI compliance options. You can add a PayPal button to help drive more sales, or use our hosted pages and offer PayPal Credit, too.

E-commerce and Online Marketing Technologies

Payment method	Payment type	Countries	Refunds
PayPal	Digital wallet	Country support	Yes
Pay Later	Loan	Australia France Germany Italy Spain United Kingdom United States	Yes
PayPal Credit	Revolving line of credit similar to a credit card	United States United Kingdom	Yes
Venmo	Digital wallet	United States	Yes
American Express	Credit card	PayPal Checkout country support , Advanced credit and debit country support	Yes
Bancontact	Bank redirect	Buyer country: Belgium Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
BLIK	Bank redirect	Buyer country: Poland Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
Discover	Credit card	PayPal Checkout country support , Advanced credit and debit country support	Yes
eps	Bank redirect	Buyer country: Austria Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days

Integration with the external systems

E-commerce and Online Marketing Technologies

giropay	Bank redirect	Buyer country: Germany Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
iDEAL	Bank redirect	Buyer country: Netherlands Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
Mastercard	Credit card	PayPal Checkout country support , Advanced credit and debit country support	Yes
MyBank	Bank redirect	Buyer country: Italy Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
Pay upon Invoice	Deferred payment	Buyer country: Germany	Within 180 days
Przelewy24	Bank redirect	Buyer country: Poland Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
SOFORT	Bank redirect	Buyer countries: Austria Belgium Germany Italy Netherlands Spain United Kingdom Merchant countries: All PayPal-supported countries except Brazil, Russia, and Japan.	Within 180 days
Visa	Credit card	PayPal Checkout country support , Advanced credit and debit country support	Yes

Integration with the external systems

E-commerce and Online Marketing Technologies

PayPal

Commercial Transaction Rates

When you accept the [User Agreement](#) to buy or sell goods or services or make any other commercial type of transaction, we call that a “commercial transaction”.

Standard rate for receiving domestic transactions

Payment Type	Rate
Alternative Payment Method (APM)	APM Transaction Rates Apply
Invoicing	3.49% + fixed fee
PayPal Checkout	3.49% + fixed fee
PayPal Guest Checkout	3.49% + fixed fee
QR code Transactions – 10.01 USD and above	1.90% + fixed fee
QR code Transactions – 10.00 USD and below	2.40% + fixed fee
QR code Transactions through third party integrator	2.29% + 0.09 USD
Pay with Venmo	3.49% + fixed fee
Send/Receive Money for Goods and Services	2.99%
Standard Credit and Debit Card Payments	2.99% + fixed fee
All Other Commercial Transactions	3.49% + fixed fee

Fixed fee for commercial transactions (based on currency received)

Currency	Fee	Currency	Fee
Australian dollar	0.59 AUD	New Taiwan dollar	14.00 TWD
Brazilian real	2.90 BRL	New Zealand dollar	0.69 NZD
Canadian dollar	0.59 CAD	Norwegian krone	3.90 NOK
Czech koruna	9.00 CZK	Philippine peso	25.00 PHP
Danish krone	2.90 DKK	Polish zloty	1.89 PLN
Euro	0.39 EUR	Russian ruble	39.00 RUB
Hong Kong dollar	3.79 HKD	Singapore dollar	0.69 SGD
Hungarian forint	149.00 HUF	Swedish krona	4.09 SEK
Israeli new shekel	1.60 ILS	Swiss franc	0.49 CHF
Japanese yen	49.00 JPY	Thai baht	15.00 THB
Malaysian ringgit	2.00 MYR	UK pounds sterling	0.39 GBP
Mexican peso	9.00 MXN	US dollar	0.49 USD

Integration with the external systems

E-commerce and Online Marketing Technologies

PayPal

Fixed fee for QR code transactions 10.01 USD and above (based on currency received)

Currency	Fee
Australian dollar	0.10 AUD
Brazilian real	0.20 BRL
Canadian dollar	0.10 CAD
Czech koruna	3.00 CZK
Danish krone	0.70 DKK
Euro	0.10 EUR
Hong Kong dollar	0.80 HKD
Hungarian forint	30.00 HUF
Israeli new shekel	0.40 ILS
Japanese yen	12.00 JPY
Malaysian ringgit	0.70 MYR
Mexican peso	2.00 MXN

Currency	Fee
New Taiwan dollar	3.00 TWD
New Zealand dollar	0.15 NZD
Norwegian krone	1.00 NOK
Philippine peso	5.00 PHP
Polish zloty	0.50 PLN
Russian ruble	3.00 RUB
Singapore dollar	0.20 SGD
Swedish krona	1.00 SEK
Swiss franc	0.10 CHF
Thai baht	4.00 THB
UK pounds sterling	0.10 GBP
US dollar	0.10 USD

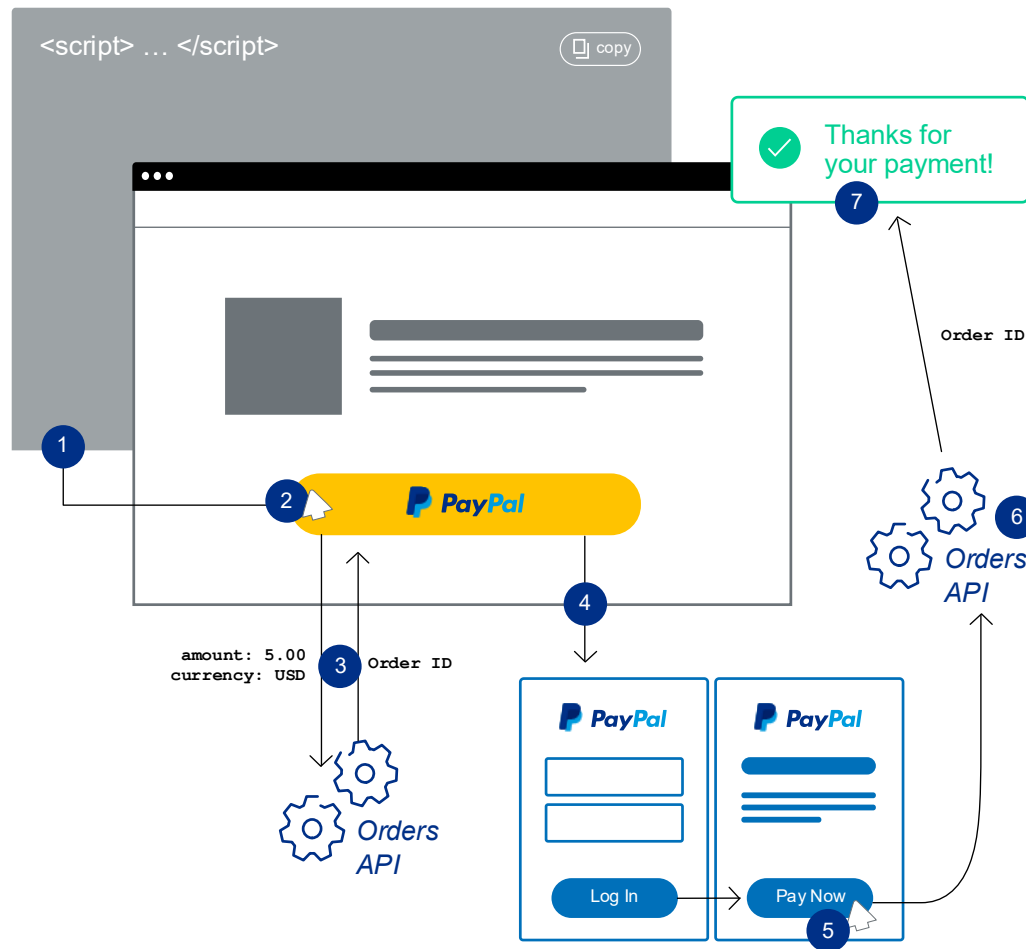
Fixed fee for QR code transactions 10.00 USD and below (based on currency received)

Currency	Fee
Australian dollar	0.05 AUD
Brazilian real	0.10 BRL
Canadian dollar	0.05 CAD
Czech koruna	1.50 CZK
Danish krone	0.35 DKK
Euro	0.05 EUR
Hong Kong dollar	0.39 HKD
Hungarian forint	15.00 HUF
Israeli new shekel	0.20 ILS
Japanese yen	6.00 JPY
Malaysian ringgit	0.20 MYR
Mexican peso	0.55 MXN

Currency	Fee
New Taiwan dollar	2.00 TWD
New Zealand dollar	0.08 NZD
Norwegian krone	0.47 NOK
Philippine peso	2.50 PHP
Polish zloty	0.23 PLN
Russian ruble	1.50 RUB
Singapore dollar	0.08 SGD
Swedish krona	0.50 SEK
Swiss franc	0.05 CHF
Thai baht	1.80 THB
UK pounds sterling	0.05 GBP
US dollar	0.05 USD

Integration with the external systems

PayPal – Flow



Integration with the external systems

Gateways – Amazon Pay – Services

Helping you do business better

We design and develop our features so you can do business better with our simple, secure and innovative services. We want to make payments easy for your team and customers.

Process Payments

Accept more payment methods globally and locally across different payment channels.

[Explore Process payments ›](#)

Installments

Enable your customers to pay equal monthly installments using their credit card and make their personal budgeting process easier.

[Explore Installments ›](#)

Collect

Get paid from anywhere online by generating a payment link that you can easily send to your customers via email or SMS.

[Explore Collect ›](#)

Monitor

Operate more effectively using real-time analytics monitoring and business intelligence features to track your operational KPIs across all your payment channels.

[Explore Monitor ›](#)

Protect

Optimize your conversions without compromising protection with our inhouse fraud experts and advanced fraud management technologies.

[Explore Protect ›](#)

Lift

Designed to better authenticate legitimate transactions and reduce friction during checkout through seamless and payment experiences.

[Explore Lift ›](#)

Amazon Pay – Accepted Methods

Visa (including the Amazon Rewards Visa Signature Card)	Amazon Store Card	Amazon Secured Card	MasterCard/EuroCard
Discover Network	American Express	Diner's Club (U.S. billing addresses only)	JCB
NYCE	STAR	China UnionPay (credit card only)	Visa, MasterCard, or American Express prepaid credit or gift cards

Gateways – Amazon Pay

Amazon Pay fees

Transaction fees

Amazon Pay fees are transaction-based and have a processing and authorization component. The transaction fee is composed of a domestic processing fee and an authorization fee, plus tax where applicable, charged when the purchase is successfully authorized and processed.

Compare a \$10.00 transaction made using different solutions for different types of goods and services:

Solution	Transaction amount	Domestic processing fee	Authorization fee: \$0.30	Total Transaction fee
Web & mobile	\$10.00	2.9% (\$0.29)	\$0.30	\$0.59
Alexa (real-world goods, services)	\$10.00	4% (\$0.40)	\$0.30	\$0.70

Gateways – Amazon Pay

Authorization fee

The \$0.30 authorization fee, included in the total transaction fee described above, will be collected when the transaction is successfully processed or when a successful authorization that has not been captured is closed or expires.

Cross-border processing fee

If you receive payment from a payment method issued outside of the U.S., we will collect a cross-border processing fee and the authorization fee described above, plus tax where applicable. The cross-border processing fee is charged only when the transaction is captured. For example, if the customer's credit card was issued outside of the U.S., the cross-border fee will be applied to your transaction.

Again, compare a \$10.00 transaction made using different solutions for different types of goods and services:

Solution	Transaction amount	Cross-border processing fee	Authorization fee: \$0.30	Total Transaction fee
Web & mobile	\$10.00	3.9% (\$0.39)	\$0.30	\$0.69
Alexa (real-world goods, services)	\$10.00	5% (\$0.50)	\$0.30	\$0.80

Gateways – Amazon Pay

Disputed chargeback fee

If you request to dispute a chargeback claim that is not covered under our Payment Protection Policy (see section B6 of the [Amazon Payments Customer Agreement](#)), we will assess a fee of \$20.00 plus tax where applicable. For more information about chargebacks, see [Handling chargebacks](#).

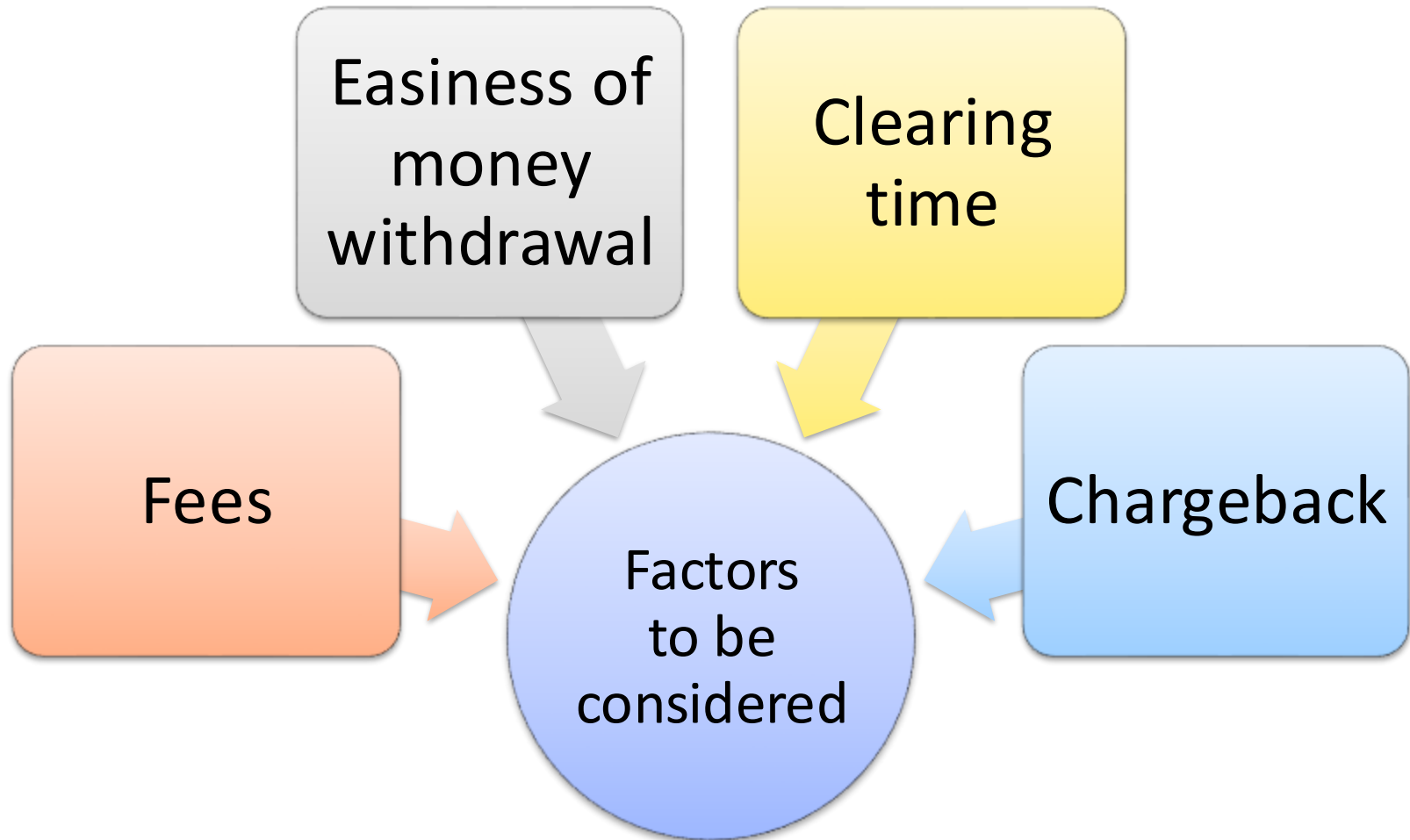
Refunds of transaction fees

If a refund takes place, you will also be refunded the following transaction fees, as applicable:

- The domestic processing fee (for example, the 2.9% fee)
- The cross-border processing fee (for example, the 3.9% fee)

Note that the Authorization fee and Disputed chargeback fee are non-refundable.

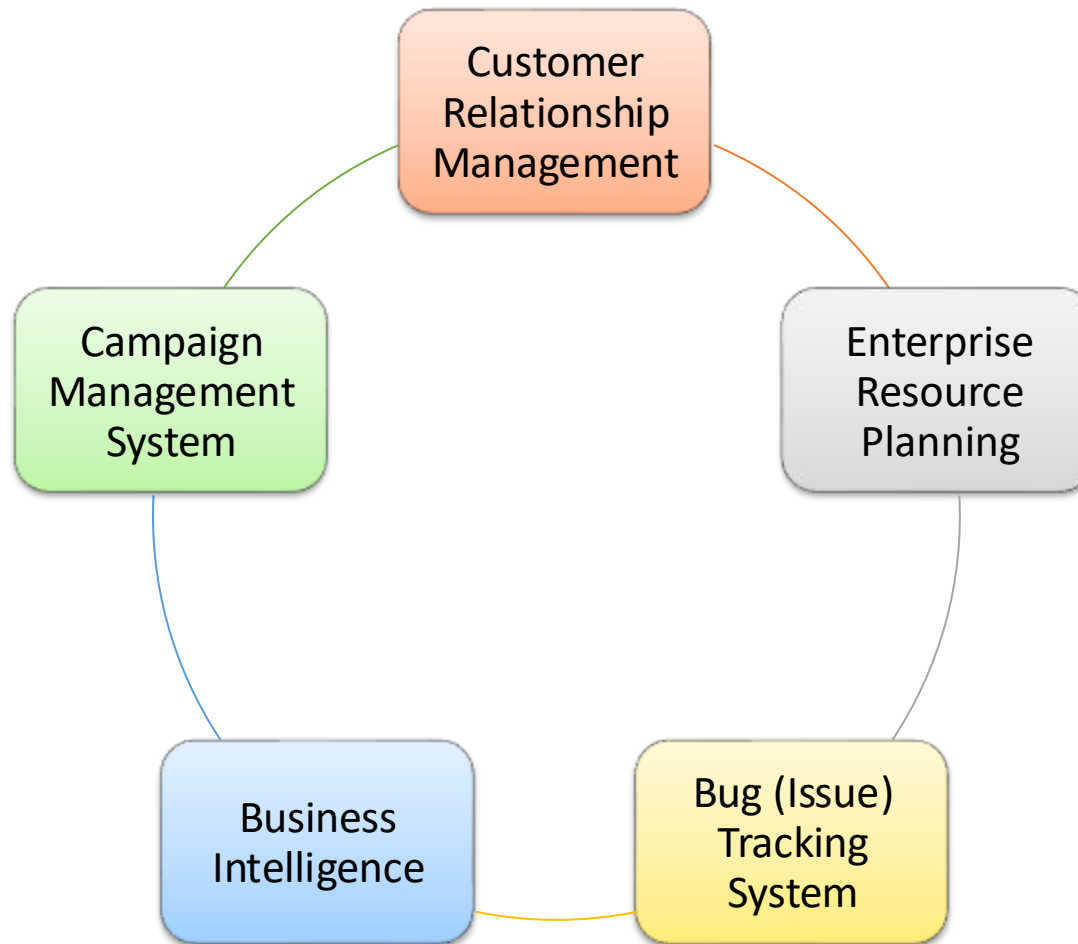
The economics of payments



INTEGRATION WITH OTHER COMMON SYSTEMS

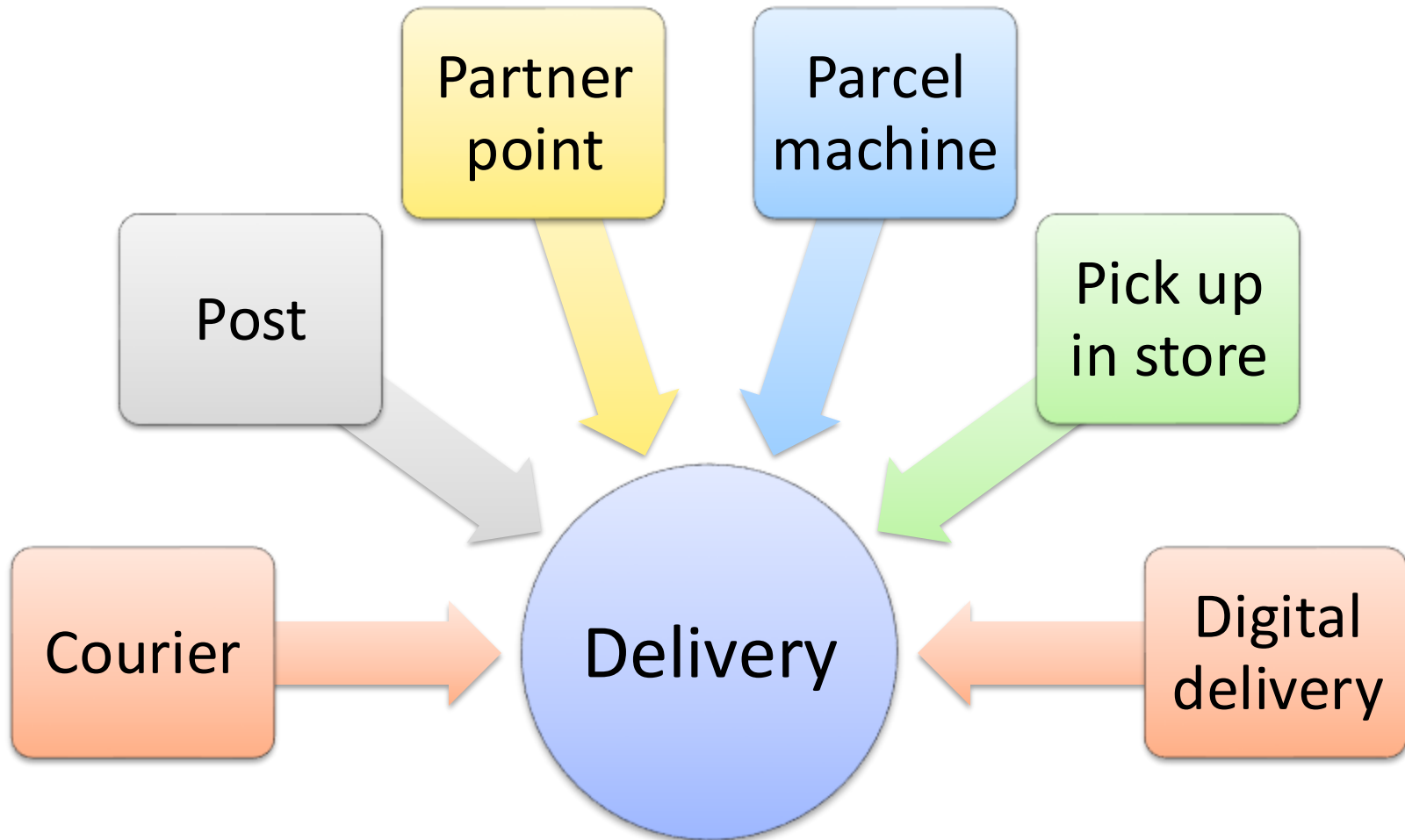
Integration with the external systems

Universal business systems



Integration with the external systems

EC-specific – Delivery



Delivery tracking



Integration with the external systems

Text searching

- A feature used to find pages/documents stored on website
- Shouldn't use a trivial searching e.g. SQL's LIKE
- This feature can be achieved by:
 - Internal CMS/F functionality
 - Integration with a controlled search engine
 - Integration with a third party search engine (Google, Bing etc.) , and searching limited to a particular domain

Text searching

Internal

- Easy to deploy
- Usually available in default installation
- Basic search capabilities
- No advanced text processing
- No additional complexity

Search engine (e.g. Elasticsearch, Solr)

- Additional element to deploy and maintain
- Full text processing pipeline
- Advanced configuration
- Advanced search capabilities
- Full control

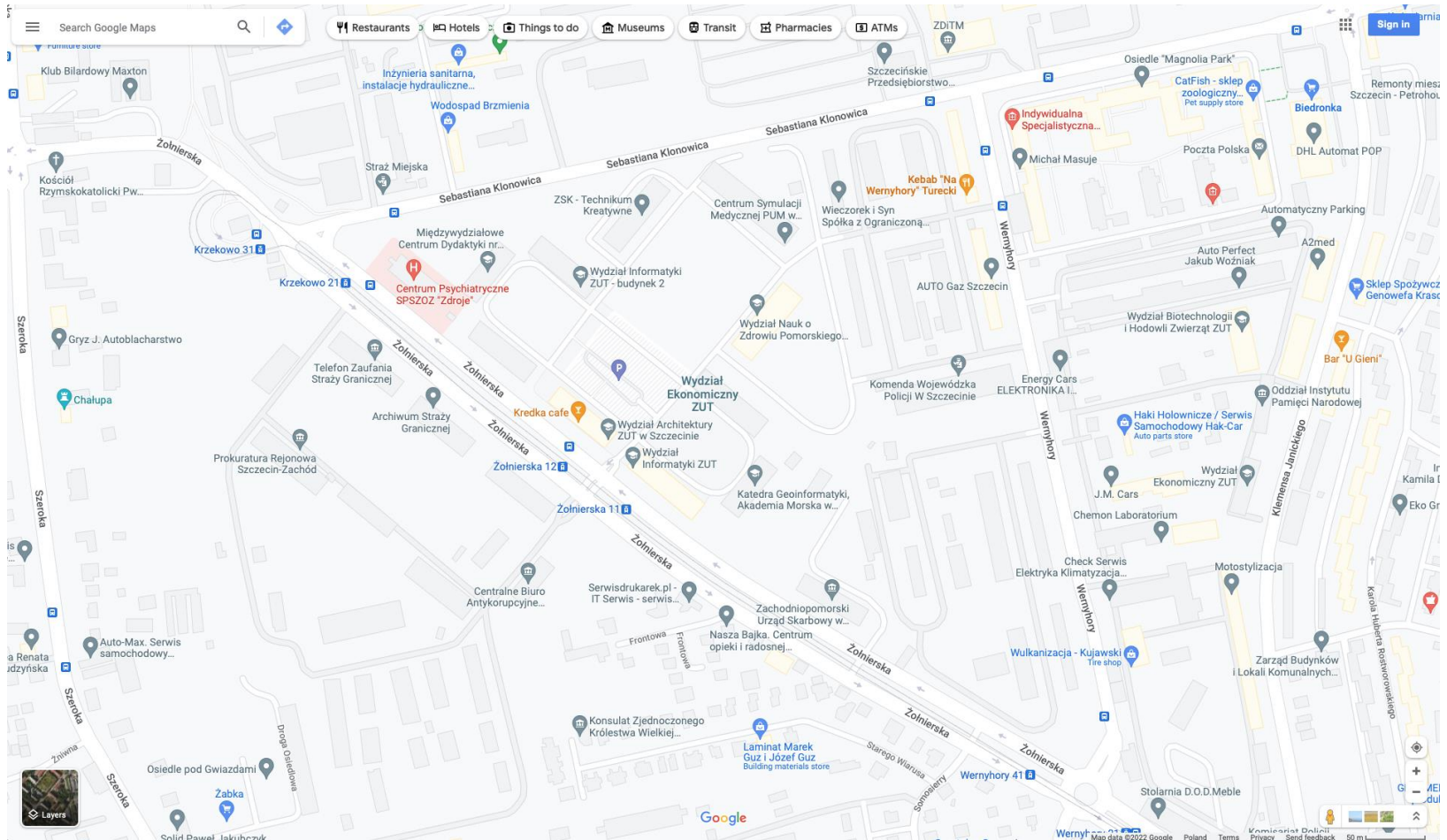
3rd party engine (Google, Bing)

- Easy to deploy
- Uncontrollable
- Full text processing pipeline
- Limited configuration
- Advanced search capabilities
- No additional elements

Mapping

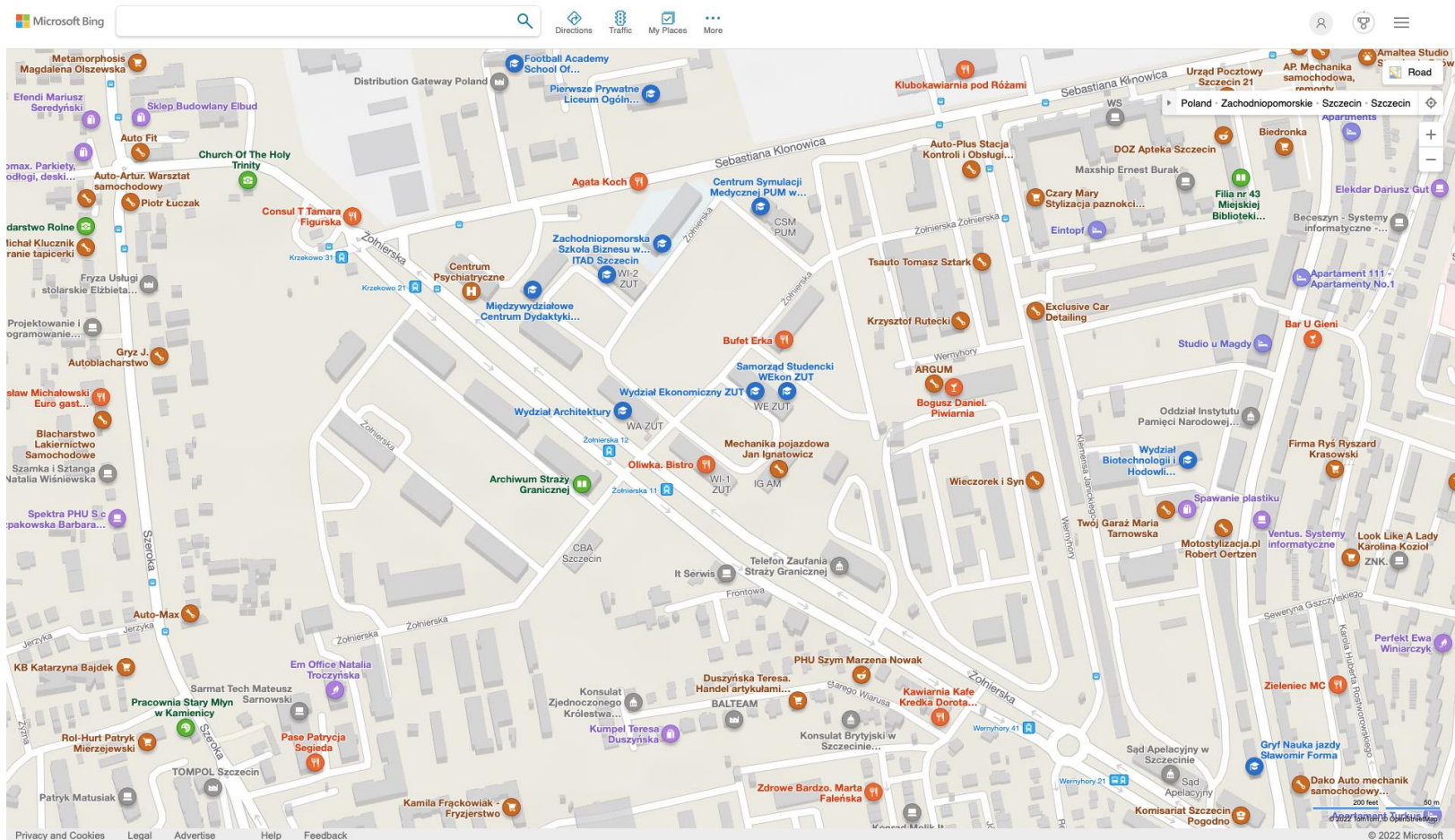
- Commonly used to deliver geographical information
- Usually provides contact information
- Very essential for specific ventures (tourist services, sport activities etc.) or companies with a lot of venues
- Gateway to additional information (photos, comments, opening hours)

Mapping – Google Maps



Integration with the external systems

Mapping – Bing Maps



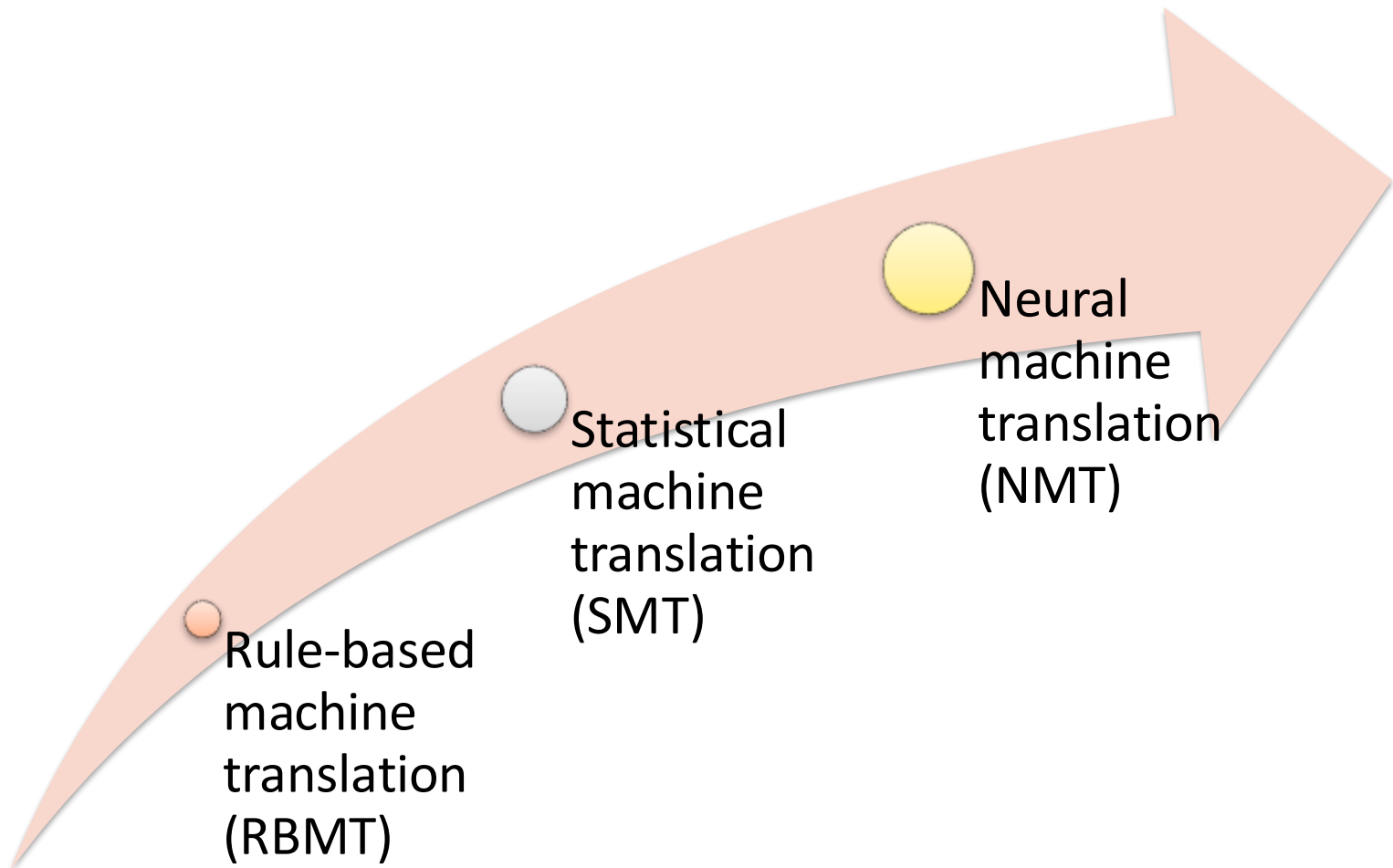
Integration with the external systems

Mapping – OpenStreetMap

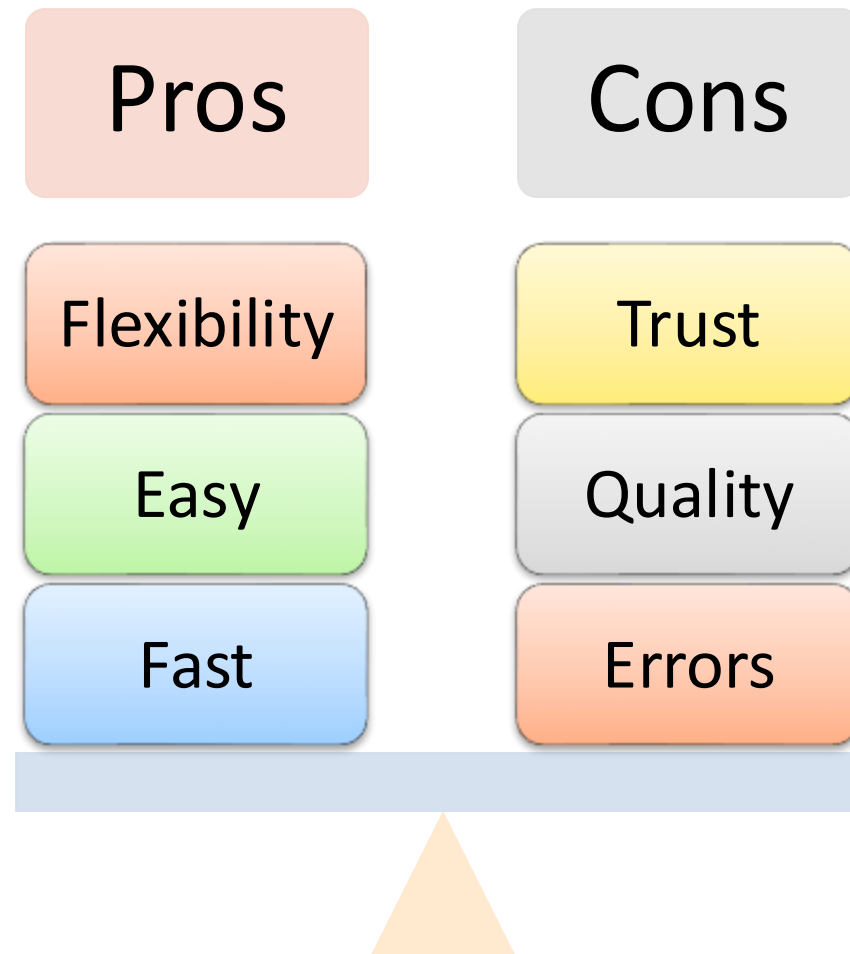


Integration with the external systems

Automated translation



Automated translation



Integration with the external systems

Automated translation

The screenshot illustrates the integration of automated translation on an e-commerce website. A language selection menu is open on the left side of the page, showing a list of languages including Polish, Spanish, English, French, German, Italian, Portuguese, Swedish, Dutch, Chinese, Polish, Korean, Japanese, Greek, Danish, Norwegian, Finnish, Catalan, and Russian. The main content area displays a product page for the Cressi Leonardo Komputer watch. The product is shown with a large image of the watch face and strap. The price is listed as PLN 745.99, and there are 145 reviews and 57 questions. A 'Dodaj do koszyka' (Add to cart) button is visible. The page also features a 'Klarna' payment option and a 'Wszystkie KATEGORIE' (All categories) link.

Automated translation – Example 1



English

Designed for professional diving, the first dive computer of this well-known Italian company has a low relief ergonomic profile to facilitate its use in all kinds of circumstances, as well as to avoid tripping and snagging in the depths.

Polish

Przeznaczony do profesjonalnego nurkowania, pierwszy komputer nurkowy tej znanej włoskiej firmy znajduje się płaskorzeźba ergonomiczny profil, aby ułatwić jego zastosowanie w różnego rodzaju okoliczności, a także aby uniknąć potknięcia i zadzierać w głębi.

Regenerated English

Dedicated to professional diving, the first dive computer of this renowned Italian company features a relief ergonomic profile to facilitate its use in all kinds of circumstances, as well as to avoid tripping and messing with the depths.

Price comparison sites

- Increases the traffic
- Compares with the competitors (use if you have competitive prices, other competitive advancements might be invisible)
- Service fee paid by the merchant
- Potential issues:
 - Test/fake sites (for systems crawling the web)
 - Matching problems (matched different items, non-matched the same items)

Price comparison engines

Google [Sign in](#)

[All](#) [Images](#) [Maps](#) [Shopping](#) [More](#)

● Dębno

Sort by: Relevance

Show only

☐ Available nearby

☐ On sale

Price

☐ Up to PLN 50

☐ PLN 50 – PLN 100

☐ PLN 100 – PLN 250

☐ PLN 250 – PLN 1,000

☐ Over PLN 1,000

PLN Min – PLN Max

Supported Network

☐ Xbox Live/Live Gold

☐ PlayStation Network/Plus

Storage Capacity

☐ 512 GB

☐ 1 TB

Colour

☐ White

☐ Black

Wireless Connectivity

☐ Wi-Fi

☐ Bluetooth

Product rating

Ads - See xbox series x

Konsola MICROSOFT XBOX Series X
PLN 2,399.00
Media Expert
★★★★★ (37,161)
Free shipping

Xbox Series X Konsola Xbox Series X
PLN 2,488.00
RTV Euro AGD
Free shipping

Konsola Xbox Series X 1TB
PLN 2,399.00
Allegro
★★★★★ (37,161)
+PLN 10.90 shipp...

Konsola Xbox Microsoft Xbox Series X
PLN 2,499.00
x-kom.pl
Free shipping

Xbox Series X + dodatkowy pad (czarny) Konsola...
PLN 2,699.00
RTV Euro AGD
Free shipping

Xbox Series X + Forza Horizon 5 Konsola Xbox...
PLN 2,595.00 P...
RTV Euro AGD
Free shipping

About these results

Microsoft Xbox x Gucci Series X Special Edition (UK Plug) Console Bundle

Konsola MICROSOFT Xbox Series X + FIFA 23
4.9 ★★★★★ 11

Konsola Xbox Series X

Konsola Xbox Series X 1TB
4.8 ★★★★★ 37,161
Backward Compatible

Price comparison engines

The screenshot displays the Ceneo website interface for the Xbox Series X. At the top, there's a navigation bar with the Sony logo and a search bar. Below this, the Ceneo logo is visible, followed by a search bar and a 'SZUKAJ' button. The main content area shows the Xbox Series X product page, including a large image of the console, a description, and a price of 2 499,00 zł. To the right, there's a 'KUP TERAZ' button and a 'DARMOWA WYSYŁKA' badge. Below the product details, there's a section for 'Najlepsze oferty wybrane na Ceneo' (Best offers selected on Ceneo), which lists several offers from different retailers like Allegro, Microsoft, Electro, and Gamefinity. Each offer includes the retailer's logo, product name, price, and a 'IDŹ DO SKLEPU' button. The bottom of the page features a 'RABATOMANIA!' banner with a 'SPRAWDŹ' button and a 'mediaexpert' logo.

SONY LEKKOŚĆ. DŹWIĘK. ANC.

CENEO Znajdź produkt lub sklep SZUKAJ

Popularne Strefa graczy PC Strefa graczy PlayStation Strefa graczy Xbox Konsole Gry planszowe Komputery

Ceneo > Gry > Konsole do gier > Microsoft Xbox Series X > **Microsoft Xbox Series X**

Xbox Series X Historia cen

Konsola do gier Microsoft Xbox Series X 1TB z 16 GB RAM z napędem optycznym BluRay. Konsola zawiera złącza: USB, HDMI. Rodzaj konsoli: Stacjonarne. Wersja: Podstawowa. Produkt zawiera: Kabel zasilający, Kabel HDMI, Pad bezprzewodowy Microsoft Xbox Series

★★★★★ 4,9 269 opinii

2 499,00 zł

KUP TERAZ

DARMOWA WYSYŁKA

dostępny

ZAMÓW Z DOSTAWĄ JUTRO!

Avans

Oferty (30) Informacje o produkcie Opinie i Recenzje (272) Pytania i odpowiedzi (12) Kup lokalnie

Najlepsze oferty wybrane na Ceneo Sortuj od najlepszych ofert Filtry

allegro ★★★★★ 31223 opinie DARMOWA DOSTAWA ALLEGRO SMART! 2 399,00 zł szczegóły dostawy dostępny IDŹ DO SKLEPU

Microsoft ★★★★★ 3 opinie Xbox Series X 2 399,00 zł DARMOWA WYSYŁKA dostępny IDŹ DO SKLEPU

ELECTRO ★★★★★ 781 opinii Konsola MICROSOFT XBOX Series X DARMOWY TRANSPORT ZAMÓW DZIŚ, ODBIERZ JUTRO 2 499,00 zł DARMOWA WYSYŁKA dostępny IDŹ DO SKLEPU

Microsoft Xbox Series X - Pozostałe oferty

gamefinity ★★★★★ 331 opinii Konsola Microsoft Xbox Series X // WYSYŁKA 24h // DOSTAWA TAKŻE W WEEKEND! // TEL. 48 660 20 30 2 389,00 zł DARMOWA WYSYŁKA dostępny IDŹ DO SKLEPU

NEONET ★★★★★ MICROSOFT Xbox Series X 1TB >> DARMOWA DOSTAWA OD 99ZŁ > SPRAWDŹ WSZYSTKIE PROMOCJE > RATY DO 2 399,00 zł

RABATOMANIA! TANIEJ Z KODEM! SPRAWDŹ mediaexpert

Recommendation engines

A separate topic – to be discusses as a part of separate lecture about recommendations

Analytics

A separate topic – to be discusses as a part of separate lecture about data-driven EC

Summary

- Integration with the external services aim at:
 - Provision of additional functionalities
 - Not wasting your time, when there are specialised supporting services (not reinvent the wheel)
 - Improving the quality of services
 - Increasing the traffic (potential customers)
 - Recommendations of other products (as mentioned to be discussed later)

TILL NEXT LECTURE